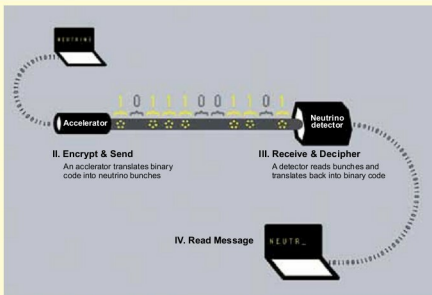


they used one of the world's most powerful particle accelerators and a multi-ton detector called MINERvA, located 100 metres under the ground. The accelerator created high-intensity beams of neutrinos by accelerating protons around a 4-kilometre-circumference track and then colliding them with a carbon target.

A message was translated into binary code. Each '1' represented neutrinos being fired, and each '0' represented those not being fired. The neutrinos were fired in large groups because they are so evasive that even with a multi-ton detector, only about one in ten billion neutrinos are detected. After the neutrinos were detected, a computer on the other end translated the binary code back into English. It is actually more complex than it sounds—systems used by the team were terribly complex, making them impractical for everyday use.

That said, this is a major breakthrough, as according to reports from the University of Rochester, this is the first neutrino experiment in the world to use a



Communicating with Neutrinos (Source: University of Rochester)

high-intensity beam for studying neutrino reactions with nuclei of five different target materials, creating the first side-by-side comparison of interactions. This will help complete the picture of neutrinos and allow data to be more clearly interpreted in current and future experiments.

6 One-atom thick material for chips

Research area: Chemistry

Applications: Sensors, integrated circuits

Researchers at NASA's Goddard Space Flight Centre have developed miniature sensors using graphene—a material comprising tightly-packed carbon atoms, available in one-atom thick sheets. Graphene is expected to be the next revolution in electronics since silicon. For the last few years, scientists have been figuring out how to manufacture devices using graphene, NASA's graphene sensors being a notable forerunner.

NASA scientists are using graphene to develop nano-sized sensors in order to detect atomic oxygen and other trace elements in the upper atmosphere and structural

strains in spacecrafts. Graphene oxidises when it absorbs atomic oxygen, producing a change in the material's electrical resistance that a graphene-based sensor could then quickly count to produce a more accurate density measurement. Graphene could also help in measuring methane, carbon monoxide and other gases. Similarly, graphene's unique properties could also help in detecting stress in materials used to make spacecraft components.

At NASA, a special technique called chemical vapour deposition (CVD) has been developed to manufacture and test a large piece of graphene. Once scientists perfect the technique to manufacture graphene, its extreme characteristics could be put to use in electronics—to make lightweight, thin, flexible and durable displays, integrated circuits, solar cells, sensors and more.

7 Navigation using Earth's magnetic field

Research areas: Geology, physics

Application: Indoor positioning systems

For many years now, scientists have been researching how birds and animals, especially turtles, navigate using changes in the Earth's magnetic field as a guideline. If quantified further, this information could be useful in designing navigational systems.

As an example, a team of Finnish researchers has de-

veloped an indoor positioning system (IPS) inspired by how homing pigeons and spiny lobsters use changes in magnetic fields to find their way. When they studied these species, they realised that some animals could sense their true position relative to their destina-



Changes in earth's magnetic field can be used as a guideline for navigation (Image courtesy: blog.willissjudd.co.uk)